

Original Article

Fructus Jujubae cooperated with water-expelling members in Shizao decoction alleviated intestinal injury and malignant ascites by modulating gut microbiota and metabolic homeostasis

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ABSTRACT

Background: Shizao decoction (SZD) consisted of *Euphorbia kansui* (EK), *Euphorbia pekinensis* (EP), *Daphne genkwa* (DG), and *Fructus Jujubae* (FJ) is a classic Chinese herbal medicine formula for treating malignant ascites, which is closely related to the modulation of gut microbiota by our previous study. For water-expelling members (WEM) including EK, EP, and DG may have side effects on the intestine, FJ is employed for detoxification and effectivity enhancement of WEM. However, the underlying mechanism for the compatibility of WEM and FJ is still unknown.

Purpose: To investigate the effect of the compatibility of WEM with FJ in SZD on malignant ascites and elucidate the potential mechanism from the perspective of the modulation of gut microbiota and related metabolic function.

Methods: Qualitative and quantitative evaluation of main components was conducted for comprehensive characterization of SZD and WEM. The effect of WEM and SZD was compared on malignant ascites effusion (MAE) rats. The intestinal injury was evaluated by HE staining and oxidative damage. Ascites weight, urine amount, fecal water content, the expression of aquaporins, and cytokines in ascites (IL-6, VEGF, and TNF- α) were measured to estimate the water-expelling activity. The intestinal flora was detected by 16S rDNA sequencing and the content of fecal short-chain fatty acids (SCFAs) was analyzed using gas chromatography-mass spectrometry. Pseudo-germ-free (PGF) and fecal bacteria transplantation animal experiments were subsequently employed to validate this finding. The fecal metabolomics and correlation analysis were finally conducted to explore the related metabolic changes.

Results: 51 and 33 components were identified in SZD and WEM, respectively. Compared to WEM alone, the compatibility with FJ remarkably reduced intestinal oxidative damage in MAE rats. Ascites was also relieved by downregulating the expression of AQP3 in the colon and decreasing the levels of IL-6, TNF- α and VEGF in ascites. The diversity of gut microbiota was reversed with an increase in *Lactobacillus* and *Clostridia* UCG-014 while a decrease in *Colidextribacter*. Under the PGF condition, compatibility of WEM with FJ failed to reduce intestinal

Abbreviations: SZD, Shizao decoction; WEM, water expelling members; EK, *Euphorbia kansui* S.L.Liou ex S.B.Ho; EP, *Euphorbia pekinensis* Rupr; DG, *Daphne genkwa* Sieb. et Zucc; FJ, *Fructus Jujubae*; MAE, malignant ascites effusion; PGF, Pseudo-germ-free; FMT, fecal bacteria transplantation; SCFAs, short-chain fatty acids; SOD, superoxide dismutase; MDA, malondialdehyde; GSH, glutathione; LDH, lactate dehydrogenase; AQP, aquaporin protein; ELISA, enzyme-linked immunosorbent assay; IL-6, Interleukin-6; TNF- α , tumor necrosis factor alpha; VEGF, vascular endothelial growth factor; GC-MS, gas chromatography-mass spectrometer; ICH, immunohistochemistry; PCoA, principal co-ordinates analysis; VIP, variable important in projection; FC, fold change; ANOVA, analysis of variance; UPLC-Q-TOF-MS, ultra high performance liquid combined mass spectrometer; 16S rDNA, 16S ribosomal RNA gene.

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injury and alleviate MA significantly, but this effect was further enhanced after FMT. 23 potential fecal metabolites were finally identified. Correlation analysis further showed that *Lactobacillus* and *Clostridia_UCG-014* were positively correlated with SCFAs and l-tryptophan. *Colidextribacter* was negatively correlated with thymidine but positively correlated with ursodeoxycholic acid and deoxycholic acid.

Conclusion: FJ cooperated with WEM reduced intestinal injury and alleviated malignant ascites by modulating gut microbiota, short-chain fatty and tryptophan metabolism. These findings provide a scientific basis for the clinical application of FJ from SZD and the safe usage of SZD.

Introduction

Malignant ascites (MA) is the abnormal accumulation of fluid in the peritoneal cavity, which is a hallmark of serious diseases, such as liver cancer, ovarian cancer, pancreatic cancer, and colorectal cancer (Ayantunde and Parsons, 2007; Cavazzoni et al., 2013). It accounts for 10 % of the ascites and becomes an implication of the end-stage tumor (Wang et al., 2023c). For MA is difficult to remove the lesion surgically, most patients suffer from debilitating symptoms including abdominal distension and pain, nausea, vomiting, fatigue and even dyspnea, considerably shortening their survival time and decreasing the quality of life (Berger et al., 2023). Currently, diuretics combined with chemotherapy may be necessarily employed to promote the excretion of fluid and simultaneously kill cancer cells (Chen et al., 2018). However, this efficacy may be weak and toxic to the wounded body. It could also exacerbate the aggregates of tumor cells to induce infections, electrolyte disturbance, and hypoproteinemia, finally increasing the generation of ascites (Ford et al., 2020). Besides, albumin is an alternative approach to decrease the ascites quickly. However, due to its limited resources, most patients could not afford the expensive price. Therefore, it is urgent to develop novel drugs for the effective treatment of MA.

Shizao decoction (SZD) is a classic formula of Chinese medicines recorded in *Shang Han Lun* and is consisted of *Euphorbia Kansui* (EK), *Euphorbia pekinensis* (EP), *Daphne genkwa* (DG) and *Fructus Jujubae* (FJ). SZD is particularly employed for the treatment of edema, ascites, and pleural effusion (Kwon et al., 2015; Zhang et al., 2021). Zhang et al. (2021) found that SZD ameliorated malignant pleural effusions and intestinal toxicity by inhibiting the release of inflammatory cytokines, promoting the expression of aquaporin, and decreasing oxidative damage. However, it needs to be paid attention that the combination of water-expelling members (WEM) including EK, EP, and DG may have the potential to induce side effects to the stomach and intestine. *Fructus Jujubae* has promising nutritional and biological effects, including anti-inflammatory, antioxidant, immunomodulatory, and anticancer activities (Fu et al., 2021; Lee et al., 2021; Periasamy et al., 2015). Recent studies indicated that compatibility with FJ was able to reduce toxicity and enhance the therapeutic effect of the major medicine. The combination of FJ with *Euphorbia fisheriana* significantly reduced liver toxicity and alleviated the cirrhotic ascites (Li et al., 2017). Oral co-administration of *Ziziphus jujuba* and *Boswellia serrata* extracts exerted anti-nociceptive and anti-inflammatory properties to inhibit the progression of osteoarthritis (Khoramjouy et al., 2022). In addition, Chinese Jujube and *Cordyceps militaris* showed a better protective effect against ethanol-induced oxidative stress by regulating Nrf2 and NF- κ B signaling pathways in zebrafish (Wang et al., 2023a). Therefore, the role of FJ in SZD may be related to protecting from intestinal injuries and enhancing the water-expelling activities of EK, EP, and DG. However, the underlying mechanism remains unclear.

Gut microbiota inhabited in the gastrointestinal tract plays a vital role in the occurrence and development of various diseases. An obvious decline in the diversity of intestinal microbiota was observed in the malignant ascites rats with an increasing abundance of *Firmicute* and *Teneriquetes* while a decreasing abundance of *Bacteroidetes* and *Proteobacteria* (Guo et al., 2021). In addition, the barrier function of the intestine was destroyed accompanied by the activation of the intrinsic immune system, leading to the release of proinflammatory cytokines

(Lakhan and Kirchgessner, 2010). At the same time, the intestinal permeability was enhanced to promote bacterial translocation, which aggravated the generation of ascites (Úbeda et al., 2016). Butyrate is one of the short-chain fatty acids fermented by intestinal microbiota and could also suppress the release of inflammatory factors (Chen et al., 2018). Acetate could increase the gut osmotic pressure to stimulate intestinal peristalsis, increasing the excretion of feces (Chen et al., 2023). These manifested that the production and excretion of ascites had close relationships with the modulation of gut microbiota. For decoctions of Chinese herbal medicines formulas mainly possess water-soluble components, and their pharmacological activities are typically related to the regulation of intestinal microbial homeostasis. Li et al. (2019) discovered that FJ improved the disorder of intestinal flora caused by toxic Crotonis Semen Pulverum to reduce small intestine injury. FJ extract may also redress the imbalanced microbial structure induced by spleen deficiency (Yi et al., 2022). FJ has abundant polysaccharides and could be responsible for immunomodulation and gastrointestinal-protective effects through modulating gut microbiota, showing prebiotic-like activities (Ji et al., 2020; Wang et al., 2023b; Zou et al., 2022). Furthermore, our previous studies illustrated that FJ in SZD could relieve the disturbed intestinal flora and the abnormal primary bile acid metabolism induced by WEM in normal rats. Hence, these results suggested that the role of FJ in SZD may be related to reducing intestinal injury and enhancing the water-expelling activity of WEM on malignant ascites by modulating the gut microbiota and metabolic homeostasis.

Therefore, in this study, the MA rat model was initially established to determine the gut microbiota related to the role of FJ in SZD. The intestinal flora was detected by 16S ribosomal RNA gene (16S rRNA) sequencing technology and the content of fecal short-chain fatty acids was analyzed using gas chromatography-mass spectrometry (GC-MS). Pseudo-germ-free (PGF) and fecal bacteria transplantation animal experiments were subsequently conducted to validate the modulation of gut microbiota in MAE rats. The fecal metabolism was finally investigated to explore the related metabolic changes.

Materials and methods

Chemicals and reagents

Euphoscopin A, 3-O-(2'E,4'Z-decadienyl)-20-O-acetylingenol (3-O-EZ), and euphol were isolated and identified by our laboratory. Rutin, quercetin, cAMP, betulinic acid, oleanolic acid, propionic acid and were purchased from Shanghai Yuanye Bio-Technology Co., Ltd (Shanghai, China). Genkwanin was purchased from Chengdu Pufei de Biotech Co., Ltd. Acetic acid, valeric acid, and butyric acid were obtained from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China) and TCI Shanghai Chemical Industrial Development Co., Ltd. (Shanghai, China). 2-methylvaleric acid was purchased from Sigma-Aldrich (Darmstadt, Germany). The purities of the above standard substances were all above 98 %. Ultrapure water was obtained by the Milli-Q system (Millipore, Bedford, MA, USA). Acetonitrile, formic acid, ethyl acetate and methanol of LC grade were purchased from Merck (Darmstadt, Germany). Ampicillin, metronidazole, vancomycin hydrochloride, and neomycin sulfate were purchased from Feiyubio Co., Ltd. (Nantong, China). Magnesium sulfate was purchased from Nanchang Baiyun Pharmaceutical Co., Ltd. (Nanchang, China). Aquaporin-2 and

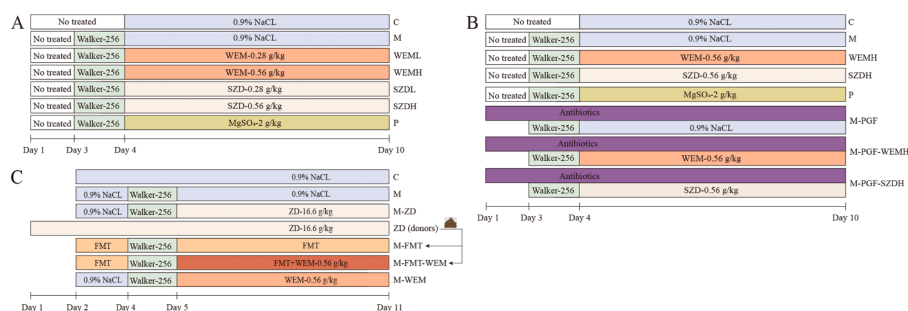


Fig. 1. Schematic mode of experimental grouping and administration in rats. (A) The mode of administration in MAE rats. (B) The mode of administration in the pseudo-germ-free MAE rats. (C) The mode of administration in the fecal-microbiota-transplantation MAE rats.

aquaporin-3 antibodies were purchased from Beijing Biosynthesis Biotechnology Co., Ltd. (Beijing, China). Malondialdehyde (MDA), glutathione (GSH), lactate dehydrogenase (LDH), and superoxide dismutase (SOD) detection kits were purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Tumor necrosis factor alpha (TNF- α), interleukin-6 (IL-6), and vascular endothelial growth factor (VEGF) enzyme-linked immunosorbent assay (ELISA) kits were purchased from Beijing Solarbio Science & Technology Co., Ltd. (Beijing, China).

Plant materials

The dried roots of EK were collected from Baoji, Shaanxi Province, China (No. NJUCM-20191015). The dried roots of EP were obtained from Bozhou, Anhui Province, China (No. NJUCM-20170616). The dried flowers of DG were collected from Anguo, Hebei Province, China (No. NJUCM-20180323). The fruits of FJ were purchased from Zhengzhou, Henan Province, China (No. NJUCM-2021042001). They were identified by Professor Yu-ping Tang (College of Pharmacy, Shaanxi University of Chinese Medicine, Xi'an, China) and the voucher specimens were stored in the Herbarium of Nanjing University of Chinese Medicine.

Preparation of WEM and SZD solutions for animal administration

DG was heated with mild fire in a hot pot until it changed the color to obtain fried DG. EK, EP, and fried DG were crushed into powders, respectively. 830 g of denucleated FJ was extracted with water twice and concentrated to 500 ml. Then, based on the principle in *Shang Han Lun*, the mixed powders of EK, EP, and fried DG powders (w/w/w, 1:1:1) were added to the water solution of FJ to obtain the high-dose (56 mg·ml⁻¹) and low-dose (28 mg·ml⁻¹) SZD solutions, respectively. Similarly, mixed powders were dispersed in the aqueous solvent to obtain high-dose (56 mg·ml⁻¹) and low-dose (28 mg·ml⁻¹) WEM suspensions (Zhang et al., 2021).

Qualitative analysis of SZD using UPLC-Q-TOF-MS

About 0.5 g of EK, EP, and fried DG powders were accurately weighed in a 100 ml conical flask. 50 ml of ethanol was accurately added and ultrasonicated for 1 h. After cooled and filtered, the supernatant was evaporated to dryness. The residue was dissolved in methanol and transferred to a 10 ml volumetric flask. This ethanol extraction was mixed equally with the aqueous solution of FJ. They were diluted twice with 80 % methanol to obtain the prepared solution of SZD. In addition, the prepared solution of WEM was acquired using water instead of FJ. The successive filtrate of these solutions was employed for qualitative analysis using UPLC-Q-TOF-MS (detailed information see the supplementary file).

Quantitative analysis of SZD

1 g of mixed EK, EP and fried DG powders (w/w/w, 1:1:1) were weighed accurately to a 100 conical flask. 50 ml of ethyl acetate was added and ultrasonicated for 60 min (power 250 W, frequency 50 kHz). After cooled and filtered, the supernatant was evaporated to dryness. The residue was dissolved in methanol and transferred to a 10 ml volumetric flask. The successive filtrate through a 0.22 μ m membrane filter was the prepared solution of WEM. In addition, the test solution of FJ was obtained using 2.8 g of freeze-dried powders of aqueous solution of FJ following the above remaining procedure. The quantitative analysis was performed after mixed standard solutions including euphoscopin A, 3-O-EZ, euphol, rutin, quercetin, genkwanin, cAMP, betulinic acid, and oleanolic acid were prepared. Determination of these 10 representative ingredients in SZD was performed using validated UPLC-based methods (detailed method see the supplementary file).

Animals and treatments

Male Sprague-Dawley rats (weight 180 ± 20 g, SPF grade) were purchased from the Qing Long Shan Dong Wu Fan Zhi Chang (Nanjing, China). All rats were kept in the experimental animal center of Nanjing University of Chinese Medicine. They were housed in autoclaved and ventilated cages under controlled conditions as follows: temperature 25 °C, humidity 50 ± 5 %, 12 h daylight cycle. The feed and water were provided sufficiently. The experiment protocols were approved by the Experimental Animal Ethics Committee of Nanjing University of Chinese Medicine (No. 202106A009, 202111A055, and 202203A083) on July 2nd, 2021, November 30th, 2021, and April 2nd, 2022.

After acclimatization, rats were randomly assigned to 7 groups ($n = 6$ per group). The normal rats were treated by oral gavage of saline solution (C group). Rats in other groups were all intraperitoneally injected with Walker 25 tumor cells to induce malignant ascites effusion (MAE) rats based on the previously established method. These MAE rats were then treated with saline solution (M group), WEM solutions at dosages of 0.28 g kg^{-1} (WEM-L group) and 0.56 g kg^{-1} (WEM-H group), SZD solutions at dosages of 0.28 g kg^{-1} (SZD-L group) and 0.56 g kg^{-1} (SZD-H group), and the magnesium sulfate solution of 2 g kg^{-1} (P group) for consecutive 7 d, respectively (Guo et al., 2021).

Then, the pseudo-germ-free (PGF) rat experiment and fecal microbiota transplantation (FMT) were conducted to validate the modulation of gut microbiota to alleviate malignant ascites and reduce intestinal injury. Some rats were randomly assigned to 8 groups ($n = 6$ per group) to verify the role of FJ in SZD on MAE rats under PGF conditions. Rats were administrated with the antibiotics' solution containing ampicillin (1 mg ml^{-1}), metronidazole (1 mg ml^{-1}), vancomycin hydrochloride (0.5 mg ml^{-1}), and neomycin sulfate (1 mg ml^{-1}) to induce PGF rats (Jiang et al., 2023; Xian et al., 2022; Zeng et al., 2020). In addition, another part of the rats was randomly assigned to 7 groups ($n = 6$ per group) to verify the synergistic effect of FJ and WEM on MAE rats treated with fecal bacteria. Rats were orally administrated with fresh fecal bacteria

suspension (1 g kg^{-1}) to induce FMT rats. The fecal bacteria suspension was supernatant after mixing fresh fecal pellets from normal rats given water solutions of FJ with sterilized PBS (w/w , 1:10) under a biological safety cabinet (Wang et al., 2018). Different modes of detailed administration are shown in Fig. 1.

The volume of ascites, feces, urine, and water content in feces

On the 4th day after administration for 4 h, rats were placed in metabolism cages and the amount of fresh feces was recorded. The fecal water content was measured after fresh samples were heated in the oven at 60°C for 24 h. The volume of urine for a consecutive 12 h was also measured. After the last administration, rats were anesthetized with 6 % sodium pentobarbital (90 mg kg^{-1}) and plasma was collected through the carotid artery. The volume of ascites was directly recorded and drawn from the abdominal cavity of MAE rats.

Spleen index

After rats were anesthetized with 6 % sodium pentobarbital, the spleen was collected and weighed. The spleen index was calculated as the ratio of spleen weight to rat body weight.

pH values of caecal content

The caecal content was weighed and mixed with five times the amount of DEPC water. The pH values of the suspensions were then determined.

Biochemical and histopathological examinations

After the last administration, rats were anesthetized with 6 % sodium pentobarbital, and plasma was collected through the carotid artery. The intestinal tissues were collected. A portion of the small intestine was homogenized, and centrifuged at 3000 g for 10 min. The supernatant was obtained for the determination of superoxide dismutase (SOD), glutathione (GSH), and malondialdehyde (MDA) based on the kit instructions.

The ileum and colon were fixed in 4 % neutral paraformaldehyde for 24 h, and then stained with hematoxylin-eosin by the Jiangsu Integrated Traditional Chinese and Western Medicine Hospital. Histopathological changes were observed under a light microscope, and three fields of each section were randomly selected.

Immunohistochemistry (IHC)

At the end of the experiment, rat kidney and colon tissues were collected and fixed in 4 % neutral paraformaldehyde for 24 h. The protein expressions of AQP3 in the rat colon and AQP2 in the rat kidney were determined through immunohistochemistry studies. Briefly, tissues were embedded in paraffin, sliced into 4–5 μm slices and

dehydrated with alcohol. The slices were then immersed in 3 % BSA for 30 min and incubated with primary AQP2 (cat. no. bs-4611R, Bioss) and AQP3 (cat. no. bs-1253R, Bioss) antibodies at 4°C overnight. HRP-labeled secondary antibody (goat anti-rabbit, Proteintech) was added for 30 min at room temperature. Then colors emerged with diaminobenzidine and the slice was stained with the hematoxylin solution again for 30 s. The density of target signals was observed and quantitatively analyzed using ImageJ software.

Enzyme-linked immunosorbent assay (ELISA)

The ascites was centrifuged at 3000 g for 10 min. The levels of IL-6, TNF- α , and VEGF in the supernatant of ascites were determined according to the specifications of ELISA kits.

16S rDNA microbial community analysis

On the 7th day after administration, fresh fecal samples (about 4–8 tablets) were collected from rats and immediately frozen in liquid nitrogen. Then fecal samples were stored at -80°C before use. The library sequencing of fecal samples was supported by OE Biotech Co., Ltd. (Shanghai, China). Briefly, total genomic DNA was extracted using MagPure Soil DNA LQ Kit (Magen, Guangzhou, China). 16S rDNA gene comprising the V3 and V4 region was amplified using the following primer pairs primer 343F: 5'-TACGGRAGGCAGCAG-3', primer 798R: 5'-AGGGTATCTAATCCT-3'. The amplicon quality was visualized using gel electrophoresis. The PCR products were purified using the AMPure XP beads (Agencourt, France) and quantified using the Qubit dsDNA assay kit (Life Technologies, United States). Sequencing was performed on an Illumina NovaSeq6000 with two paired-end read cycles of 250 bases each.

Quantitative determination of short-chain fatty acids

200 mg of fecal samples were accurately weighed in a 2 ml centrifuge tube and 1 ml of water was added. The mixture was then homogenized and centrifuged at 13,000 g for 10 min. 500 μl of the supernatant was mixed with 20 μl of the internal solution (2-methylvaleric acid) and 20 μl of 50 % hydrochloric acid for 20 s. Then 570 μl of ethanol was added and vortexed extracted for 3 min. The supernatant was employed to determine the short-chain fatty acids (SCFAs) including acetic acid, propionic acid, butyric acid, and valeric acid using the established gas chromatography-mass spectrometry (GC-MS) (Zhu et al., 2022) (detailed information in the supplementary file, Tables S1 and S2).

Fecal metabolomics analysis

After the administration on the 5th day, rats were placed in the metabolism cages for feces collection in 12 h. 100 mg of feces was weighed accurately and 500 μl of water was added. The mixture was homogenized for 5 min and then centrifuged at 13,000 g for 10 min. 400

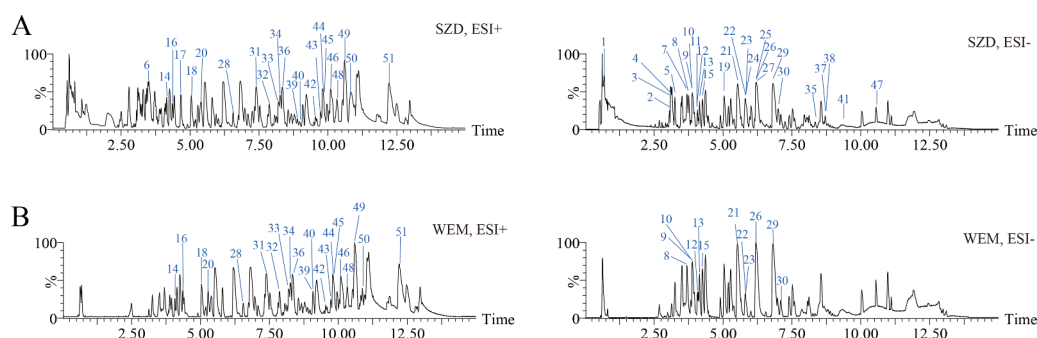


Fig. 2. Representative total ion chromatograms of SZD (A) and WEM (B) extracts in both positive and negative ion modes using UPLC-Q-TOF-MS/MS.

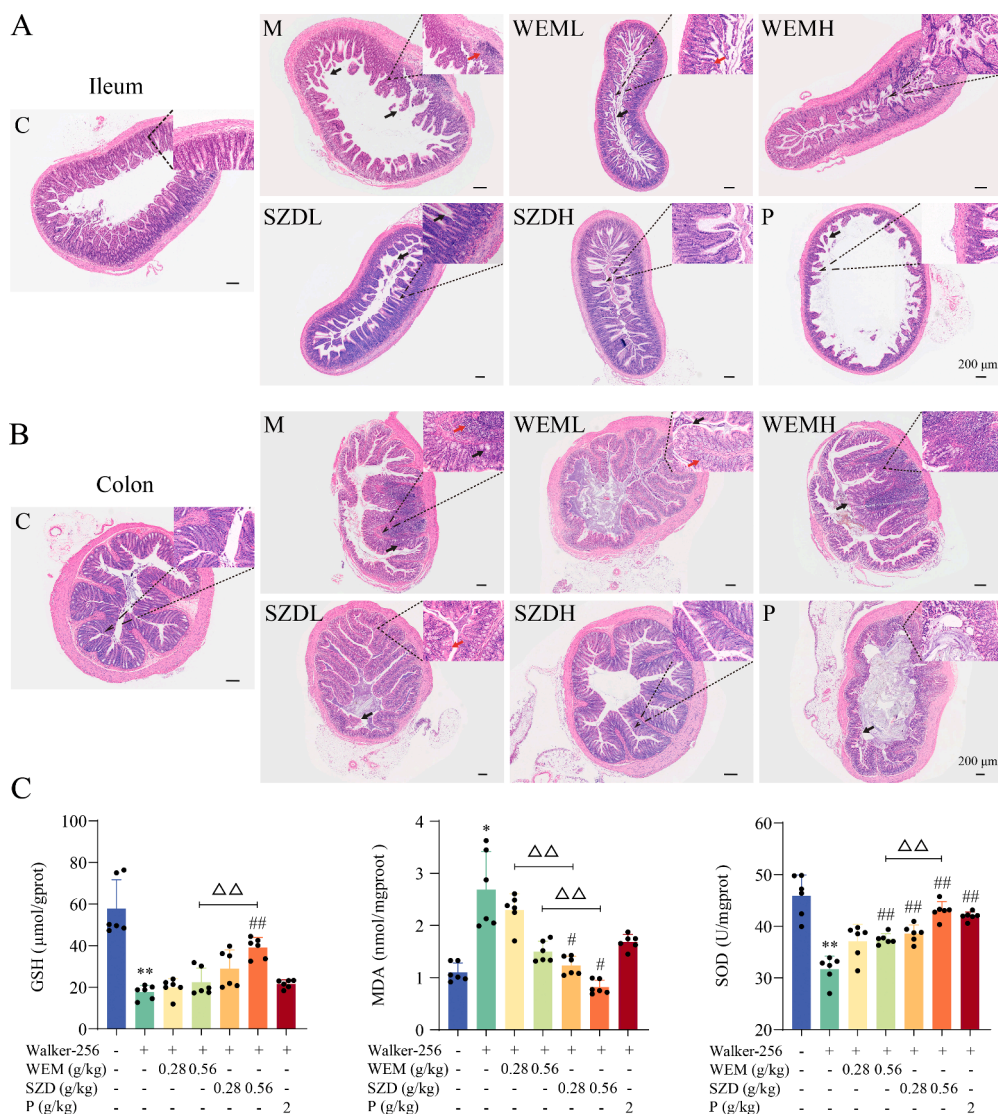


Fig. 3. FJ and WEM synergistically alleviated the intestinal injury in malignant ascites effusion (MAE) rats ($n = 6$). (A) Representative histopathological images of rat ileum tissues (black arrow, villous lesions; red arrow, inflammatory infiltrate). (B) Representative histopathological images of rat colon tissues. (C) The levels of GSH, MDA and SOD in rat ileum. C, control group; M, model group; WEML, MAE rats administrated with toxic members in SZD at the dosage of $28 \text{ mg}\cdot\text{ml}^{-1}$; WEMH, MAE rats administrated with toxic members in SZD at the dosage of $56 \text{ mg}\cdot\text{ml}^{-1}$; SZDL, MAE rats administrated with SZD at the dosage of $28 \text{ mg}\cdot\text{ml}^{-1}$; SZDH, MAE rats administrated with SZD at the dosage of $56 \text{ mg}\cdot\text{ml}^{-1}$; P, MAE rats administrated with magnesium sulfate at the dosage of $2 \text{ g}\cdot\text{kg}^{-1}$. * $p < 0.05$ vs the C group; ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; Δ represents $p < 0.05$; $\Delta\Delta$ represents $p < 0.01$.

μl of the supernatant was transferred to another tube as the extraction solution of the first step. Next, $500 \mu\text{l}$ of methanol was added to the precipitate. The mixture was homogenized for 5 min, and centrifuged at $13,000 \text{ g}$ for 10 min. $400 \mu\text{l}$ of the supernatant was mixed with the extraction solution of the first step. The final solution was filtered through a $0.22 \mu\text{m}$ microporous membrane for LC-MS analysis.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 8.0. For data that followed normally distributed and equal variance, a one-way analysis of variance (ANOVA) followed by the least significant difference t -test was conducted to describe differences among experimental groups. Otherwise, a non-parametric Kruskal-Wallis test combined with the Wilcoxon test was used for comparisons among groups. P -values < 0.05 were considered statistically significant.

Results

Qualitative and quantitative analysis of chemical components in SZD

The qualitative and quantitative analysis of SZD was initially conducted to give a comprehensive quality characterization of SZD. The chemical components in SZD were characterized using UPLC-Q-TOF-MS. The total ion chromatograms of SZD and WEM for both positive and negative modes were shown in Fig. 2. 51 components were identified in SZD and detailed information including retention time (t_R), molecular weight and mass fragment ions (m/z) of these compounds were seen in Table S3. Thereinto, 33 compounds originated from WEM and 22 ingredients in SZD were introduced by FJ. The main constituents that accounted for a large proportion were terpenes, flavonoids, benzyl methanol glycosides, and cAMP. Then, the contents of 9 representative components containing euphoscopin A, 3-O-EZ, euphol, rutin, quercetin, genkwanin, cAMP (FJ), betulinic acid (FJ), and oleanolic acid (FJ) were determined based on the validated UPLC methods (detailed information see Tables S4–S7). Thereinto, cAMP serves as a signaling

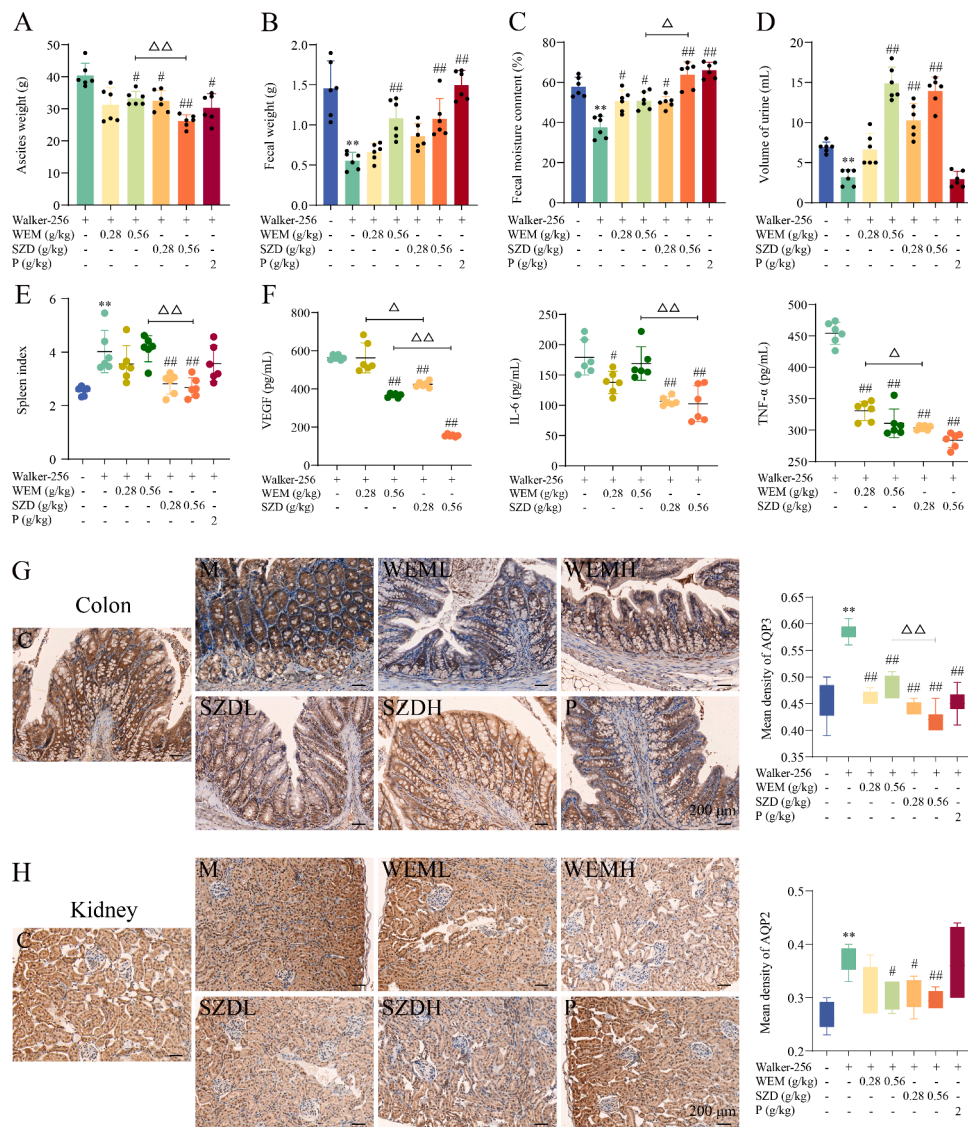


Fig. 4. FJ enhanced the water-expelling activity of WEM in SZD on the malignant ascites ($n = 6$). (A)–(E) The ascites weight, fecal weight, fecal moisture content, the volume of urine, and the spleen index of rats in all groups. (F) Contents of VEGF, IL-6, and TNF- α in rat ascites. (G)–(H) Representative immunohistochemical images and statistical analysis of aquaporin 3 and aquaporin 2 in the rat colon and kidney, respectively. ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; Δ represents $p < 0.05$; $\Delta\Delta$ represents $p < 0.01$.

molecule that is related to intestinal inflammation (Li et al., 2022). Betulinic acid improved the intestinal physical and immunological barrier by enhancing the expression of zonula occludin 1 and Claudin-1 (Wang et al., 2019). Oleanolic acid treatment could improve the disorder of intestinal flora and the metabolic bile acid spectrum (Wang et al., 2022). The results were 6.240, 44.51, 1428, 210.1, 24.31, 434.8, 337.6, 166.3, and 116.2 $\mu\text{g}\cdot\text{g}^{-1}$, respectively.

FJ and WEM synergistically alleviated the intestinal injury in MAE rats

For intestinal tract injuries may be occurred in MAE, we assumed that SZD and WEM could ameliorate the intestinal injury in MAE rats. The results of HE staining of the colon and ileum tissues in rats from different groups were shown in Fig. 3A and B. In malignant ascites rats, intestinal structures including the epithelial mucosae and propria mucosae were seriously damaged accompanied by sparse goblet cells localized in the epithelial mucosae layer. However, after the administration of WEM, these conditions were alleviated. More importantly, in comparison with the WEM group, a distinct reduction in MAE rats' intestinal injuries was seen for SZD groups. Oxidative damage of the ileum

was further evaluated using GSH, MDA, and SOD (Fig. 3C). WEM and MgSO_4 merely increased the level of SOD in MAE rats ($p < 0.01$). However, under the existence of FJ, the levels of GSH and SOD were significantly increased with the decreased level of MDA in groups with SZD treatment, especially in the high-dose group ($p < 0.05$). More importantly, the levels of GSH, MDA, and SOD showed better improvement in the SZD group than those in the WEM group ($p < 0.01$). These data indicated that FJ cooperated with WEM synergistically reduced intestinal injury in MAE rat.

FJ enhanced the water-expelling activity of WEM in SZD on the malignant ascites

FJ was then considered to enhance the water-expelling activity of WEM on malignant ascites rats. In Fig. 4A–D, ascites was significantly generated in MAE rats. Compared to the C group, the fecal moisture content and the amount of urine were significantly decreased in the M group ($p < 0.01$). WEM administration ($0.56 \text{ g}\cdot\text{kg}^{-1}$) reduced the ascites, increased the water content of feces, and promoted the excretion of urine ($p < 0.05$). After the combination of WEM and FJ, SZD showed

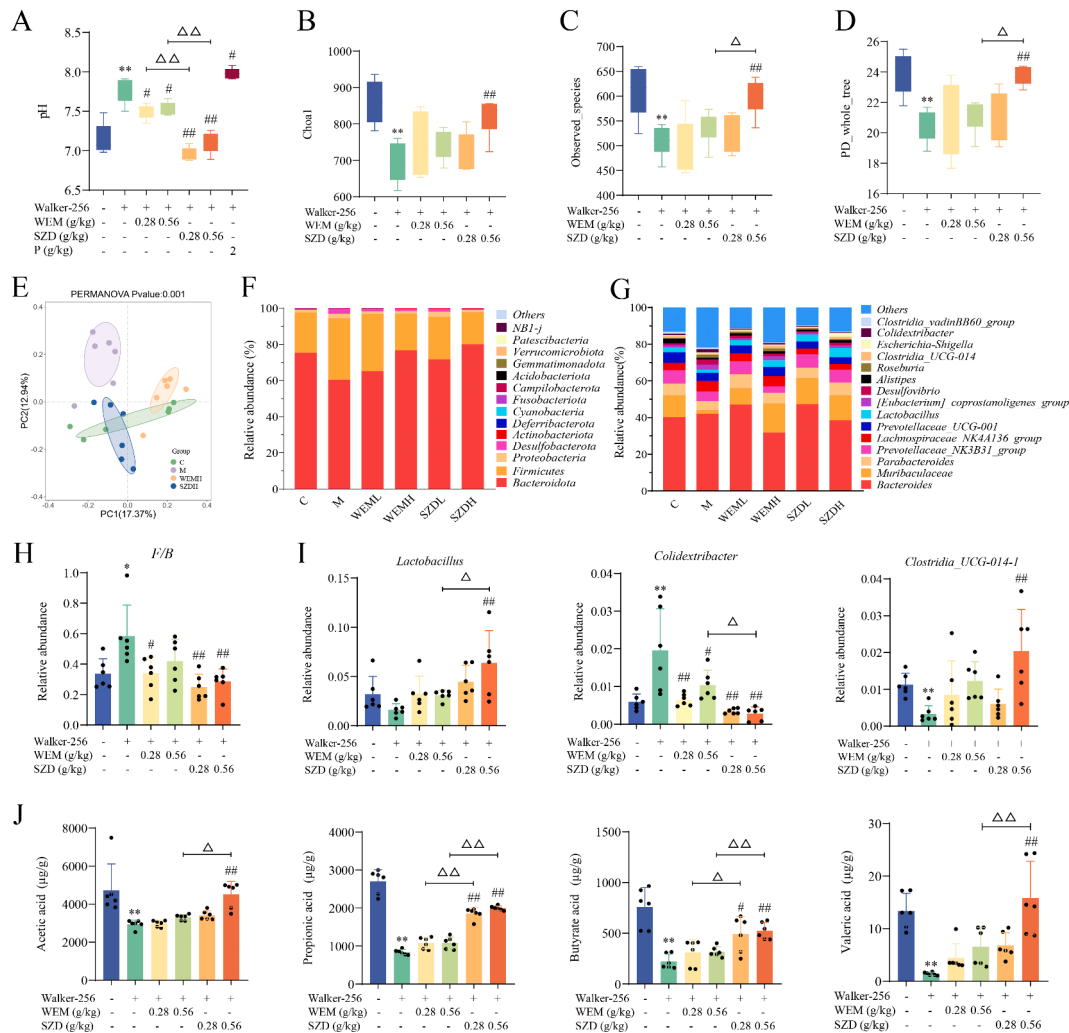


Fig. 5. FJ cooperated with WEM reversed the intestinal flora disorder and the loss of beneficial SCFAs production in MAE rats ($n = 6$). (A) pH values of rat feces. (B)–(D) The alpha diversity analysis of gut microbiota including Chao1, observed-species and PD whole tree index. (E) PCoA score plot of β -diversity analysis for gut microbiota. (F)–(G) The relative abundance of the gut microbial composition at the phylum and genus levels. (H) The ratio of the relative abundance of *Firmicutes* to *Bacteroidetes*. (I) The differential relative abundance of *Lactobacillus*, *Colidextribacter*, and *Clostridia* UCG-014-1 between the C and M groups at the genus level. (J) The contents of SCFAs including acetic acid, propionic acid, butyrate acid, and valeric acid in rat feces using GC-MS. * $p < 0.05$ vs the C group; ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; Δ represents $p < 0.05$; $\Delta\Delta$ represents $p < 0.01$.

better improvement on the above indicators compared to the WEM group at the same dosage ($p < 0.01$). However, MgSO₄ mainly decreased the ascites by increasing the fecal weight and moisture content ($p < 0.05$). Because the water-exPELLing activity of WEM was related to the modulation of the microenvironment of ascites and the expression of aquaporins (AQPs), the pro-inflammatory cytokines in ascites and the expression of AQP3 and AQP2 were also detected (Fig. 4E–H). Compared with the C group, the spleen index was significantly increased in MAE rats ($p < 0.01$), which indicated there was inflammation under the ascites condition. High levels of VEGF, IL-6, and TNF- α in the rat ascites confirmed this assumption. Meanwhile, AQP3 and AQP2 were highly expressed in the rat colon and kidney, respectively ($p < 0.01$). Only SZD administration decreased the abnormal spleen index of MAE rats ($p < 0.01$). Both WEM (0.56 g·kg⁻¹) and SZD (0.28 and 0.56 g·kg⁻¹) decreased the levels of the pro-inflammatory factors and inhibited the protein expressions of AQP3 and AQP2 ($p < 0.05$). Moreover, compared to the WEM group, the levels of IL-6, TNF- α , and VEGF in ascites were much decreased and the expression of AQP3 in the colon was significantly suppressed after SZD administration at the dosage of 0.56 g·kg⁻¹ ($p < 0.01$). These results manifested that FJ enhanced the water-exPELLing activities of WEM in SZD to reduce the

malignant ascites by increasing the fecal water content and urine, decreasing the levels of VEGF, IL-6, and TNF- α in ascites, and inhibiting the protein expression of AQP3 in the colon.

Compatibility of WEM with FJ reversed intestinal flora disorder and the loss of beneficial SCFAs production in MAE rats

For the intestinal microbiome played a key role in the production and excretion of ascites, we supposed that FJ cooperated with WEM in SZD alleviated the malignant ascites through the alternation of gut microbiota. In Fig. 5A, compared with the C group, the pH values of the intestinal content were significantly increased in MAE rats ($p < 0.01$). A distinct decline was observed after administration with WEM and SZD ($p < 0.05$). In addition, the abnormal pH values were reversed better with SZD administration than with WEM ($p < 0.01$). These data also suggested that the effect of SZD may be related to the modulation of the gut microbiota. The bacterial diversity and richness of fecal samples were initially evaluated using the α -diversity and β -diversity analysis. The α -diversity was measured by calculating the Chao1 index, observed-species, and PD whole tree index (Fig. 5B–D). A significant decrease in these indicators occurred in MAE rats ($p < 0.01$). These indexes were

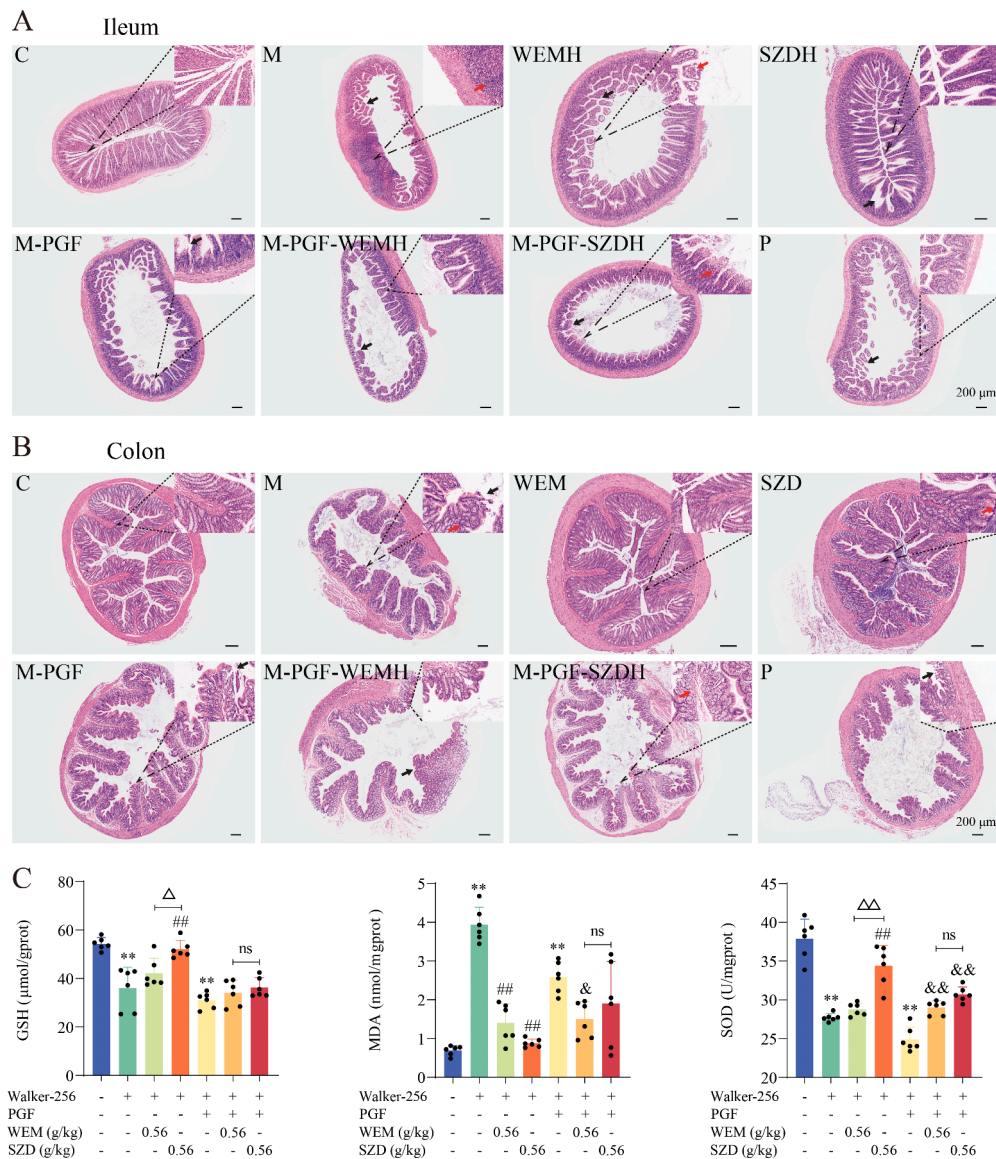


Fig. 6. FJ failed to cooperate with WEM to alleviate the intestinal injury in pseudo-germ-free MAE rats ($n = 6$). (A)–(B) Representative histopathological images of the rat ileum and colon tissues in the PGF rat experiment. (C) The levels of GSH, MDA, and SOD of the rat ileum in the PGF rat experiment. $^{**}p < 0.01$ vs the C group; $^{\#}p < 0.05$ vs the M group; $^{\&}p < 0.01$ vs the M group; $^{\&\&}p < 0.05$ vs the M-PGF group; $^{\Delta}p < 0.05$; $^{\Delta\Delta}p < 0.01$; ns represents no significance.

merely reversed with SZD administration ($0.56 \text{ g}\cdot\text{kg}^{-1}$, $p < 0.01$). Thereinto, compared with the WEM-H group, observed-species and PD_whole_tree index were significantly increased in the SZDH group ($p < 0.05$). Then, the bacterial community compositions of different groups were investigated using β -diversity analysis (Figs. 5E, S1). The C and M groups were separated spatially. WEM and SZD groups were located between the C and M groups, indicating that WEM and SZD were adjusting the abnormal state of the MAE rat to the regular condition. Specifically, SZD groups were much closer to the C group than WEM groups at both dosages of $0.28 \text{ g}\cdot\text{kg}^{-1}$ and $0.56 \text{ g}\cdot\text{kg}^{-1}$. Subsequently, the microbial structures were explored at phylum and genus levels (Fig. 5F–I). At the phylum level, *Bacteroidetes*, *Firmicutes*, *Protobacteria*, *Desulfobacterota* accounted for over 99 %. The ratio of the relative abundance of *Firmicutes* to *Bacteroidetes* was observably increased but was decreased after WEM and SZD administration ($p < 0.05$). There was a tendency for improvement with SZD compared to WEM. At the genus classification, a higher relative abundance of *Colidextribacter* ($p < 0.05$) and a lower relative abundance of *Lactobacillus* and *Clostridia_UCG-014* ($p < 0.05$) were seen in MAE rats than those in normal rats. While,

compared with the M group, evident reversion could be noticed after the treatment with SZD ($0.56 \text{ g}\cdot\text{kg}^{-1}$, $p < 0.05$). More importantly, compared to the WEM group, there was a significant increase in *Lactobacillus* with a decrease in *Colidextribacter* for the SZD group ($p < 0.05$). Finally, four SCFAs produced by gut microbiota including acetic acid, propionic acid, butyric acid, and valeric acid in feces were determined using GC–MS (Fig. 5J). Their levels were significantly decreased in MAE rats ($p < 0.01$). This condition was reversed with SZD treatment ($p < 0.05$). Compared to WEM groups, SZD groups exhibited higher concentrations of 4 SCFAs ($p < 0.05$). The above results suggested that the compatibility of WEM with FJ in SZD reversed the intestinal microbial disorder and increased the beneficial short-chain fatty acids in MAE rats.

FJ failed to cooperate with WEM to alleviate intestinal injury and malignant ascites in MAE–PGF rats

To confirm whether the compatibility of WEM with FJ alleviated the intestinal damage and malignant ascites relied on the intestinal microbiota, PGF rat experiments were conducted. Because FJ cooperated with

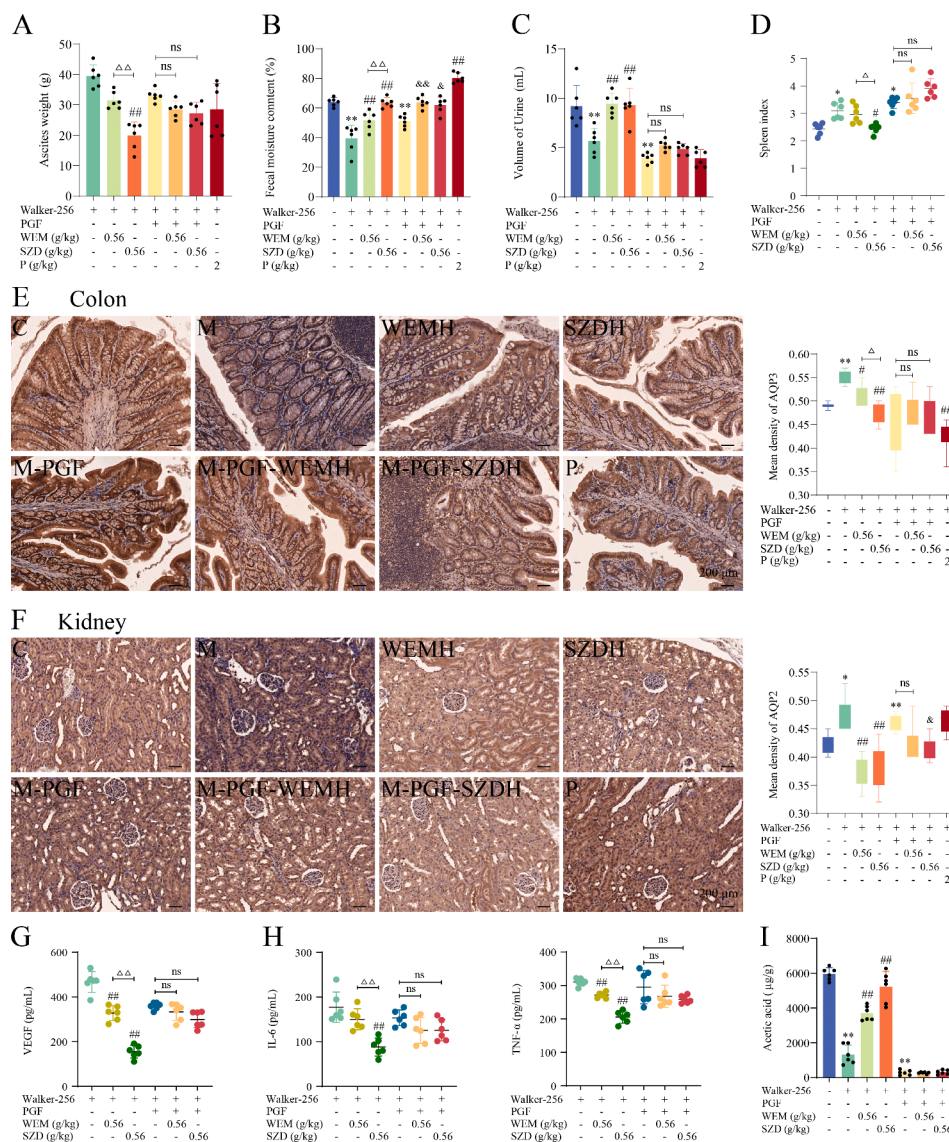


Fig. 7. FJ failed to cooperate with WEM to reduce malignant ascites in PGF-MAE rats ($n = 6$). (A)-(D) The ascites weight, fecal moisture content, the volume of urine, and the spleen index of rats in the PGF rat experiment. (E)-(F) Representative immunohistochemical images and statistical analysis of aquaporin 3 and aquaporin 2 in the rat colon and kidney in the PGF rat experiment, respectively. (G)-(H) The levels of VEGF, IL-6, and TNF- α in rat ascites in the PGF rat experiment. (I) The fecal content of acetic acid in the PGF rat experiment. * $p < 0.05$ vs the C group; ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; & $p < 0.05$ vs the M-PGF group; && $p < 0.01$ vs the M-PGF group; ns represents no significance.

WEMs synergistically alleviated the intestinal injury and FJ enhanced the water-expelling activity of WEM against malignant ascites through the modulation of gut microbiota in a dose-independent manner. Considering the animal welfare, high-dosage groups of WEM and SZD were selected for validation experiments. HE staining results showed that shrunken glands and shortened intestinal villus were observed in MAE-PGF rats (Fig. 6A, B). Meanwhile, the level of MDA was increased and that of GSH and SOD were decreased significantly ($p < 0.01$, Fig. 6C). But, the length of the intestinal villus was not reversed with WEM and SZD treatment. SZD failed to ameliorate the intestinal injury concerning SOD, GSH and MDA better than WEM in MAE rats under PGF condition.

The water-expelling activity was then investigated on MAE rats under the PGF condition. Ascites was produced with significantly decreased fecal water content and the amount of urine in MAE-PGF rats ($p < 0.01$). AQP2 protein was highly expressed in the kidney and the spleen index was enhanced ($p < 0.05$). High levels of VEGF, IL-6, and TNF- α were also observed (Fig. 7A-H). WEM and SZD administration

failed to reverse this condition, except for an increase in the fecal moisture content. However, there was no significant difference between the WEM group and SZD group under the PGF circumstance. In addition, only acetic acid was detected in rat feces. Because most gut microbiota was eliminated by antibiotics, there was a low-level of acetic acid in the MAE-PGF rats' feces. This could not be improved by WEM and SZD treatment (Fig. 7I). These data demonstrated that FJ could not enhance the water-expelling activity of WEM on malignant ascites under the PGF conditions. All the above results illustrated that the compatibility of WEM with FJ to reduce intestinal damage and alleviate malignant ascites relied on the intestinal flora.

Fecal transplantation from FJ-administrated rats cooperated with WEM further reduced intestinal injury and malignant ascites

The fecal microbiota transplantation (FMT) experiment was further conducted to validate the role of FJ to enhance the pharmacological activities of WEM on malignant ascites by regulating gut microbiota. In

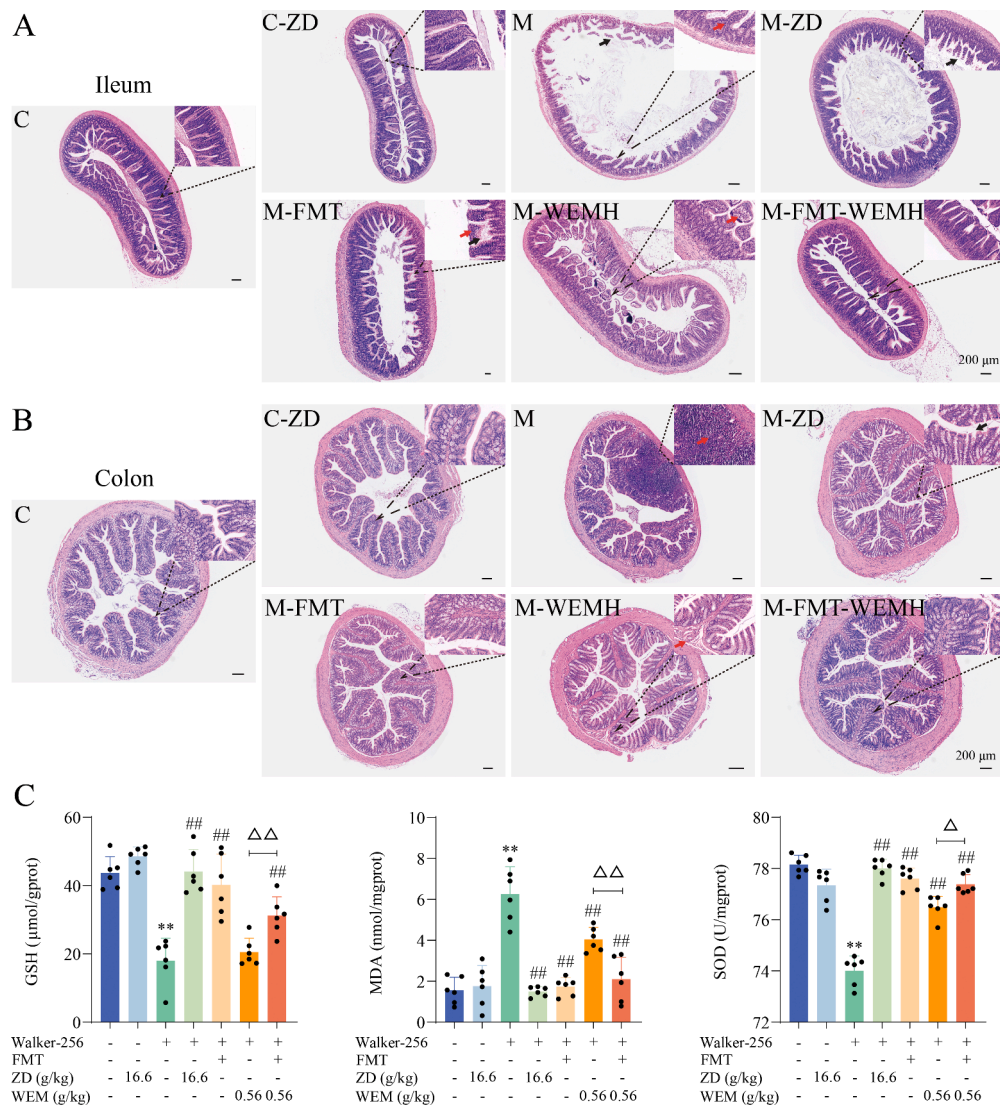


Fig. 8. Fecal transplantation from FJ-administrated rats cooperated with WEM further reduced intestinal injury ($n = 6$). (A)-(B) Representative histopathological images of the rat ileum and colon tissues in the FMT rat experiment. (C) The levels of GSH, MDA, and SOD in the rat ileum in the FMT rat experiment. C-ZD, normal rats administrated with FJ decoction at the dosage of $16.6 \text{ g} \cdot \text{kg}^{-1}$; M-ZD, MAE rats administrated with FJ decoction at the dosage of $16.6 \text{ g} \cdot \text{kg}^{-1}$; M-FMT, MAE rats with fecal transplantation from FJ-fed rats. ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; Δ represents $p < 0.05$; $\Delta\Delta$ represents $p < 0.01$. (ZD, *Ziziphus jujuba* Mill decoction; FMT, fecal microbiota transplantation).

Fig. 8A, B, FJ alleviated the inflammatory injuries of MAE rats and administration of fecal bacteria from FJ-fed rats lengthened intestinal villus. Compared to the M group, the levels of GSH and SOD were significantly increased with the declined level of MDA ($p < 0.01$), indicating that FJ and FJ-regulated gut microbiota could ameliorate the intestinal injuries in MAE rats (Fig. 8C). More importantly, compared to the M-WEMH group, a distinct improvement was observed after co-administrated with WEM and fecal microbiota derived from the FJ-fed rats ($p < 0.05$). The water-exPELLing activities were subsequently investigated. In Fig. 9, compared with the M group, FJ and FMT reduced the ascites ($p < 0.05$), increased fecal water content ($p < 0.01$), down-regulated the protein expression of AQP3 in the colon ($p < 0.01$), and decreased the level of VEGF, IL-6, and TNF- α in ascites ($p < 0.01$). Moreover, after transplanted with the FJ-fed rats' microbiota, the above indicators showed better improvements than WEM alone ($p < 0.05$). However, FJ and FMT failed to increase the amounts of feces and urine through the modulation of AQP2. These results showed that FJ may alleviate malignant ascites and decrease intestinal injury. More importantly, FJ-fed microbiota could synergistically reduce intestinal injury and enhance the water-exPELLing activities of WEM on the MAE rats in

two main ways. One is to inhibit the expression of intestinal AQP3 to promote the excretion of water in feces. The other is to reduce the levels of VEGF, IL-6, and TNF- α to decrease the ascites production.

Fecal transplantation from FJ-administrated rats cooperated with WEM regulated the intestinal flora disorder and reversed the loss of beneficial SCFA production in MAE rats

In Fig. 10A, FJ and fecal microbiota derived from FJ-fed rats decreased the intestinal pH values of MAE rats ($p < 0.05$). Besides, there was a significant difference between the M-WEMH and M-FMT-WEMH groups ($p < 0.01$). The bacterial communities were further evaluated for different groups. After the administration with FJ and FJ-fed rats' fecal microbiota, bacterial diversity and richness showed an increase concerning the Observed-specie and Shannon indexes when compared to the MAE rats ($p < 0.01$, Fig. 10B, C). Compared with the WEM treatment alone, these indicators exhibited an increasing trend in the combination of WEM and FMT. The bacterial community composition was then discussed using β -diversity analysis (Fig. 10D). All the administration groups located between the C and M groups, which were adjacent to the

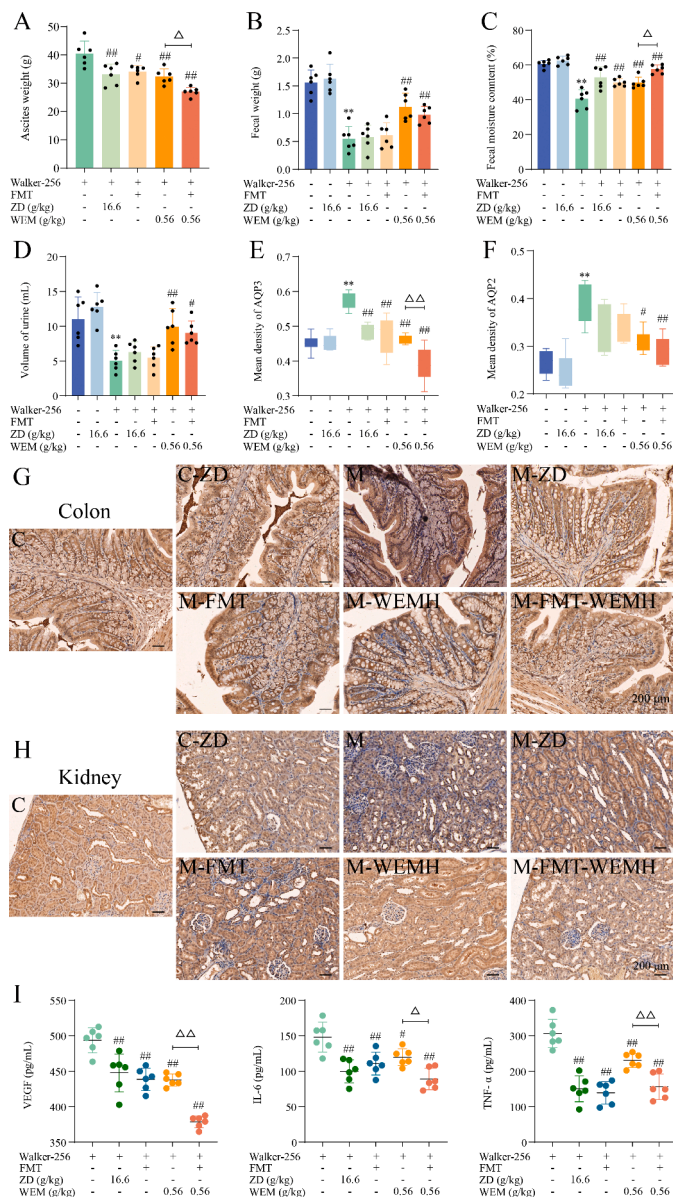


Fig. 9. Fecal transplantation from FJ-administrated rats further enhanced the water-exPELLing activity of WEM on malignant ascites ($n = 6$). (A)–(D) The ascites weight, fecal weight, fecal moisture content, and the volume of urine in the FMT rat experiment. (E)–(H) The statistical analysis and representative immunohistochemical images of aquaporin 3 and aquaporin 2 in rat colon and kidney in the FMT rat experiment, respectively. (I) The levels of VEGF, IL-6, and TNF- α in rat ascites in the FMT rat experiment. ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; Δ represents $p < 0.05$; ΔΔ represents $p < 0.01$.

C group. Thereinto, the FMT-WEMH group was much closer to the C group. Fig. 10E further showed the differences in the abundance of various taxa among all groups. FJ feces treatment with WEM significantly increased the ratio of *Firmicutes* to *Bacteroidetes* compared to the WEM treatment alone ($p < 0.05$). At the genus level, FJ feces treatment with WEM remarkably improved the relative abundance of *Prevotellaceae-NK3B31-group*, *Lactobacillus*, and *Lachnospirillum-NK4A136-group* ($p < 0.05$), while decreased that of *Escherichia-shigella* ($p < 0.05$, Fig. 10F, G). FJ decoction modulated the relative abundance of three genera including *Prevotellaceae-NK3B31-group*, *Lactobacillus*, and *Escherichia-shigella* ($p < 0.05$). Specifically, the relative abundance of *Lactobacillus* was distinctly enhanced after co-administration of FJ feces and WEM compared to WEM alone ($p < 0.01$). Finally, results of

the determination of microbiota-fermented fecal SCFAs showed that FJ and FJ-fed microbiota significantly increased the levels of 4 SCFAs in MAE rats ($p < 0.05$, Fig. 10H). In addition, their contents were much higher in FJ feces treatment combined with WEM than those in WEM treatment alone ($p < 0.05$). These results showed FJ regulated gut microbiota to ameliorate malignant ascites. More importantly, fecal bacteria from FJ-administrated rats could enhance the water-exPELLing activity and decrease intestinal injury of WEM, which may be related to the increase of *Lactobacillus* and improvement of SCFAs.

Compatibility of WEM with FJ reversed the metabolic function that was associated with the modulation of gut microbiota in MAE rats

The fecal metabolomics strategy was finally applied to analyze the metabolic reversion that was associated with the modulation of gut microbiota of compatibility of WEM with FJ. Representative total ion chromatograms in both positive and negative modes were displayed in the supplementary files (Fig. S2). Principle component analysis was employed to observe the distribution of fecal samples from different groups. The M group located apart from the C group. WEMH and SZDH groups were distributed between the C and M groups, indicating that WEM and SZD treatment were reversing the abnormal state. Thereinto, the SZDH group was closer to the C group than the WEMH group with the addition of FJ (Figs. 11A and S3). Then, the partial least squares-discriminant analysis (PLS-DA) model between the C group and the M group was built. The PLS-DA score plot exhibited that a distinct separation was seen between these two groups (Figs. 11B and S4). The permutation test ($n = 200$) confirmed that the model fit well (Figs. 11C and S5). The differential metabolites between the C group and M group were identified based on the VIP, p -value, and fold change. Combining the significant differences between the SZDH and WEMH groups, 23 potential metabolites were screened out (Table S8). Compared with the C group, 13 differential metabolites were decreased significantly while others were increased in the M group ($p < 0.05$). The recovery was observed in both WEM and SZD groups, and SZD showed better modulation on these metabolites than WEM ($p < 0.05$). Then, 23 metabolites were imported to MetaboAnalyst for pathway enrichment (Fig. 11D). They were mainly involved in the linolenic acid metabolism, histidine metabolism, tryptophan metabolism, and tyrosine metabolism with impact values of 1.00, 0.22, 0.14, and 0.10, respectively.

To further understand the interactions between gut microbiota and fecal metabolites, spearman's correlation analysis was conducted to elucidate the potential relationship. Results between 15 gut bacteria and 28 fecal metabolites (including 24 differential metabolites and 4 SCFAs) were shown in Fig. 11E. Compared with the WEMH group, SZD significantly increased the relative abundance of *Lactobacillus* and *Clostridia-UCG-014*, which were positively correlated with acetic acid, butyrate acid, valeric acid, propionic acid, l-tryptophan, and l-histidine, but negatively correlated with linoleic acid. In addition, SZD remarkably decreased the relative abundance of *Colidextribacter*, which was negatively correlated with thymidine and positively correlated with 2-hydroxystearic acid, ursodeoxycholic acid, and deoxycholic acid.

Discussion

SZD is a classic formula of Chinese medicines recorded in *Shang Han Lun*. SZD has excellent water-exPELLing activity against malignant ascites, which are contributed by main compositions including EK, EP, and DG. Previous studies showed that SZD alleviated the malignant pleural effusions through promoting the expression of intestinal AQP8 and inhibiting renin-angiotensin-aldosterone system. But the severe side effects to the liver and gastrointestinal tract of EK and EP seriously restricted their clinical usage. Hence, FJ in this formula is to decrease the side effects and further enhance the effectivity. Compatibility with FJ relieved the intestinal injury caused by EK, EP, and DG in normal rats through restoring disordered intestinal flora and related metabolic

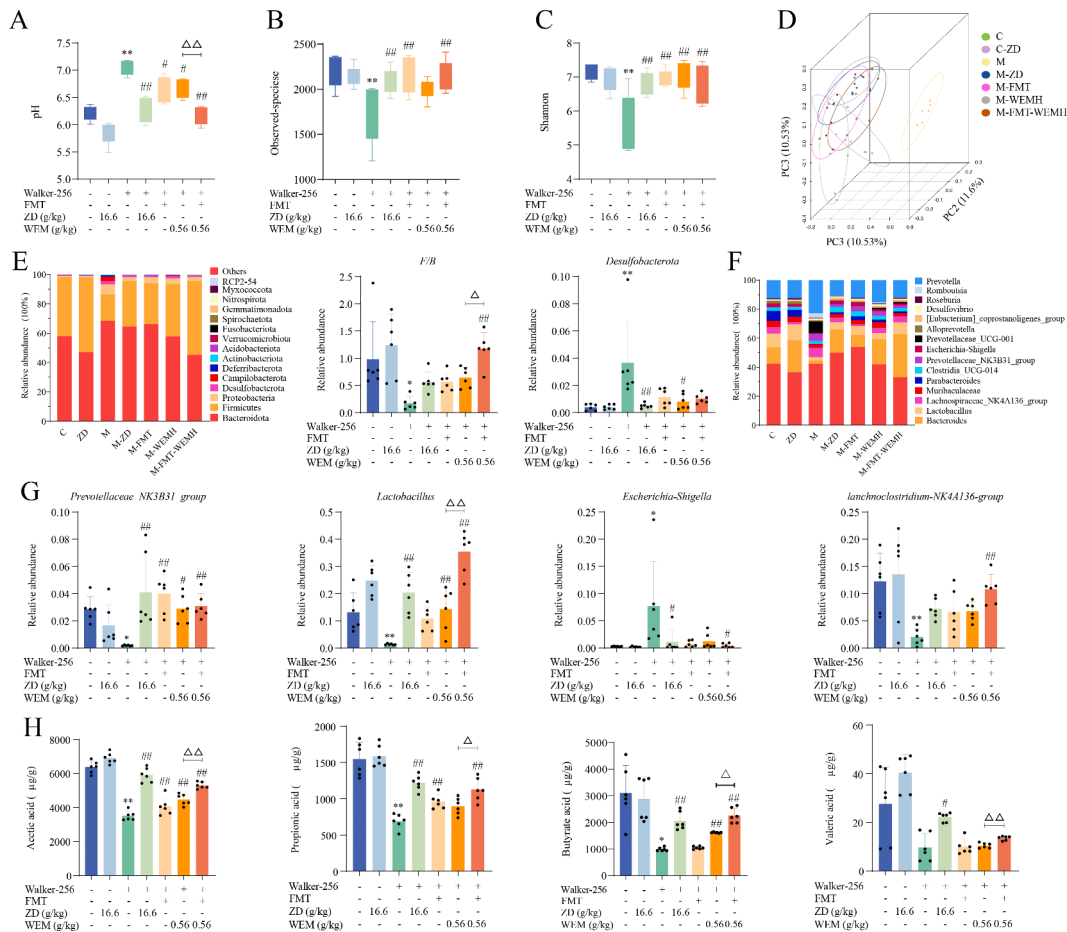


Fig. 10. Fecal transplantation from FJ-administrated rats cooperated with WEM regulated the intestinal flora disorder and reversed the loss of beneficial SCFA production in MAE rats ($n = 6$). (A) pH values of rat feces. (B)–(C) The alpha-diversity analysis of gut microbiota including observed-species and Shannon index. (D) PCoA score plot of β -diversity analysis for gut microbiota in the FMT rat experiment. (E) The relative abundance of the gut microbial composition at the phylum level. The ratio of the relative abundance of *Firmicutes* to *Bacteroidetes* and the relative abundance of *Desulfobacterota*. (F) The relative abundance of the gut microbial composition at the genus level. (G) The differential relative abundance of *Prevotellaceae_NK3B31_group*, *Lactobacillus*, *Escherichia-Shigella*, and *Lachnospiraceae_NK4A136_group* between the C and M groups. (H) The contents of acetic acid, propionic acid, butyrate acid, and valeric acid in rat feces in the FMT rat experiment. * $p < 0.05$ vs the C group; ** $p < 0.01$ vs the C group; # $p < 0.05$ vs the M group; ## $p < 0.01$ vs the M group; Δ represents $p < 0.05$; $\Delta\Delta$ represents $p < 0.01$.

pathways. Therefore, in this study, the role of FJ in SZD to reduce intestinal injury and ameliorate malignant ascites was explored from the perspective of the modulation of gut microbiota and metabolic homeostasis.

Previous exploration of our group showed that SZD had promising activity against malignant effusion at the dosages of $0.28 \text{ g}\cdot\text{kg}^{-1}$ and $0.56 \text{ g}\cdot\text{kg}^{-1}$ (twice and four times clinical dosage, respectively). At these dosages, no obvious intestinal injury was observed in malignant effusion rats. Hence, in this study, $0.28 \text{ g}\cdot\text{kg}^{-1}$ and $0.56 \text{ g}\cdot\text{kg}^{-1}$ for low and high dosages were adopted with oral gavage. Recent studies indicated that AQP2 and AQP3 in the kidney and colon played an important role in the regulation of fluid metabolism when ascites occurred (Ando, 2021; Silberstein et al., 1999). In addition, ascite fluid contains a range of endothelial and inflammatory cells (Ford et al., 2020). The pro-inflammatory cytokines including interleukin-6 (IL-6) and interleukin-1 β , tumor necrosis factor (TNF- α), and vascular endothelial growth factor (VEGF) were highly increased, which accelerated the formation of malignant ascites (Yin et al., 2016). For example, VEGF promoted tumor angiogenesis, permeabilization, and infiltration to induce the simulation of liquid digestion to increase the peritoneal permeability, leading to the accumulation of ascites. Such malignant ascites fluid resulted in the occurrence of peritoneal carcinomatosis that negatively influenced surgical respectability (Wei et al., 2017). Our data showed that compared to the usage of water-expelling members alone,

compatibility with FJ significantly decreased the ascites formation by increasing the fecal water content. This was related to the inhibition of the expression of AQP3 in the colon and the remarkable decline in the levels of IL-6, TNF- α and VEGF in ascites. In addition, the combination of FJ and water-expelling members better relieved the intestinal injury by protecting it from oxidative damage.

CHM formulas are commonly given oral administration, which may interact with the gut microbiota to exert their pharmacological activities (Lin et al., 2021). The microbial composition changes are related to the development of various gastrointestinal diseases. In addition, treatment with microbial agents has the potential to shrink tumors (Qiu et al., 2022). The feces were sequenced by 16S RNA sequencing and the microbial composition was analyzed. In addition, SCFAs were fermented by gut microbiota and served as vital endogenous substances for *in-vivo* metabolic processes. Acetate could increase the gut osmotic pressure to stimulate the intestinal peristalsis, increasing the excretion of feces. Propionate decreased the content of serum cholesterol and reduced the carcinogenic effects (Jiang et al., 2020). Butyrate is a source of metabolic energy and an inhibitor of histone deacetylase, which helps to maintain the integrity of the intestine, inhibits the release of pro-inflammatory factors, and promotes the death of tumor cells (Zhan et al., 2022). The results showed that the balance of gut microbiota was destroyed with decreased microbial diversity for MAE rats. The relative abundance of *Lactobacillus* and *Clostridia-UCG-014* was decreased but

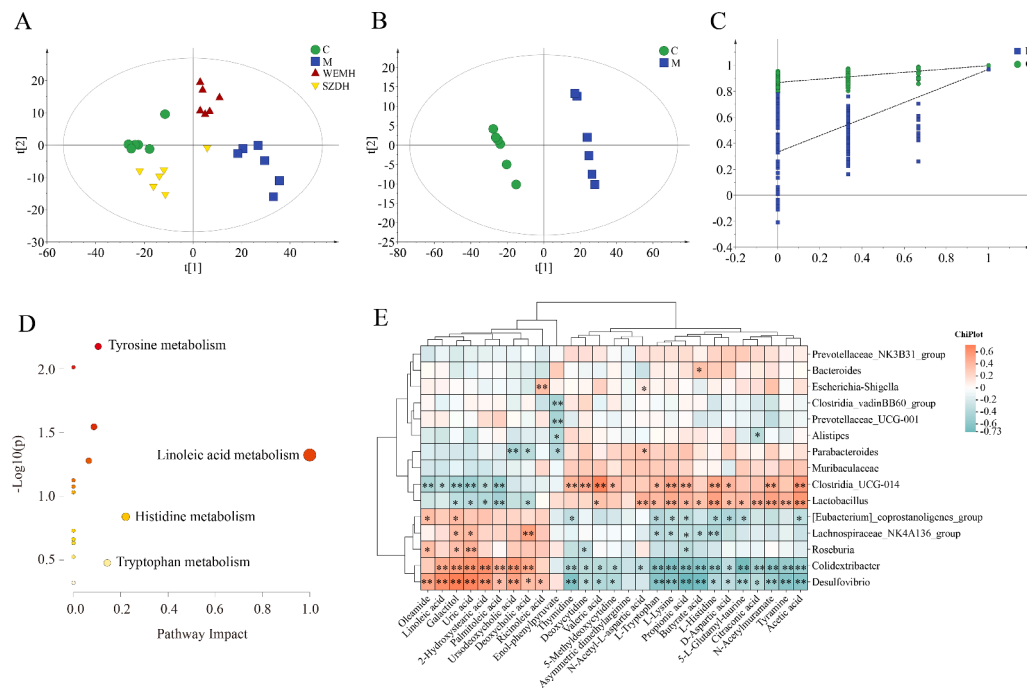


Fig. 11. The role of FJ in SZD altered the metabolites of feces in MAE rats ($n = 6$). (A) PCA score plot among C, M, WEMH, and SZDH groups. (B) The score plot of the established PLS-DA model between C and M groups. (C) Permutation test ($n = 200$) of the PLS-DA model. (D) Pathway analysis based on the differential fecal metabolites. (E) Spearman's correlation analysis between the gut bacteria and differential fecal metabolites. * means $p < 0.05$, ** means $p < 0.01$.

that of *Colidextribacter* was increased. *Lactobacillus* participates in carbohydrate metabolism and amino acid metabolism to exert promising activity against malignant ascites (Zhang et al., 2020). *Lactobacillus paracasei* subsp. *paracasei* CNCM I-1518 improved the gut barrier and reduced pro-inflammatory factors to decrease bacterial translocation in colitis (Sánchez et al., 2017). norank_F_norank_O_Clostridia_UCG-014 could balance the microbial composition between beneficial and harmful bacteria to restore intestinal flora and enhance immunity (Liu et al., 2021). Moreover, the colonized *Clostridia-UCG-014* may have the potential to ameliorate colitis and commensal bacterial damage (He et al., 2022). *Colidextribacter* could induce oxidative stress and damage the gut barrier function, which leads to harmful substances entering the blood (Leibovitzh et al., 2022). Meanwhile, an obvious decline in the contents of 4 SCFAs including acetic acid, propionic acid, butyrate acid, and valeric acid was observed. Compared to the usage of water-expelling members alone, compatibility with FJ remarkably further increased the microbial diversity and reversed the alternation of these differential microbiota and SCFAs. These manifested the role of FJ to reduce intestinal injury and enhance the water-expelling activity, which was related to the reverse of intestinal flora disorder and beneficial SCFAs decline.

To confirm this correlation, MAE-PGF rats and fecal transplantation from FJ-administrated rats were employed to validate the role of FJ and gut microbiota. Antibiotics possess broad-spectrum antimicrobial activities and are used to eliminate the gut microbiota, which causes numerous gastrointestinal side effects (Reyman et al., 2022). The results declared that the MAE-PGF rats were successfully established because SCFAs were not detected except for acetic acid. Ascites was formed accompanied by severe intestinal injury. This may be because antibiotics damaged the intestinal barrier and produced endogenous ligands such as lipopolysaccharide to enter the systemic circulation, which interacted with TLR4 to stimulate inflammatory responses (Tang et al., 2021; Wu et al., 2020). Under this condition, compared to the usage of water-expelling members alone, compatibility with FJ failed to further reduce intestinal injury and reduced ascites of MAE rats. Then, fecal microbiota transplantation (FMT) derived from FJ-fed rats to MAE rats was conducted to further validate the mechanism. Currently, FMT has

become an alternative approach to treating gastrointestinal diseases, such as inflammatory bowel disease and Crohn's disease, due to the modulation of intestinal microbial dysbiosis (Shao et al., 2021; Sokol et al., 2020). The data illustrated that administration of FJ and FJ-fed microbiota reduced intestinal injury and alleviated malignant ascites to some extent. Compared to the application of water-expelling members alone, compatibility with fecal microbiota from FJ-fed rats significantly suppressed the formation of ascites by increasing the fecal water content, which was associated with an increase in the expression of AQP3 and a decrease in the levels of pro-inflammatory cytokines in ascites. In addition, the diversity of gut microbiota was remarkably enhanced accompanied by the regulation of the balance between differential pathological and beneficial genera. Interestingly, a significant increase in the contents of SCFAs for ZD treatment also demonstrated that FJ was beneficial to the modulation of gut microbiota, which was consistent with the results of the literature (Han et al., 2022). This also indicated that the compatibility of FJ with WEM in SZD was closely related to the modulation of gut microbiota, which further confirmed our assumption. Meanwhile, there was no significance for the contents of SCFAs between ZD group (the green column) and FMT+WEM group (the red column). It emphasized the role of FJ in the modulation of gut microbiota for alleviating malignant ascites. These validation experiments demonstrated that FJ cooperated with water-expelling members reduced intestinal injury and alleviated malignant ascites through modulating gut microbiota.

The metabolic alternation affected by gut microbiota was then investigated using fecal metabolomics. Combining the differential characterizations between the pathological changes and the role of FJ in SZD, 23 potential metabolites were screened that were mainly involved in the linolenic acid metabolism, histidine metabolism, tryptophan metabolism, and tyrosine metabolism. Correlation analysis was finally conducted to explore the interaction between the gut microbiota and host metabolism. The results indicated that compared to the usage of water-expelling members alone, compatibility with FJ significantly increased the relative abundance of *Lactobacillus* and *Clostridia-UCG-014*, which were positively correlated with acetic acid, butyrate acid,

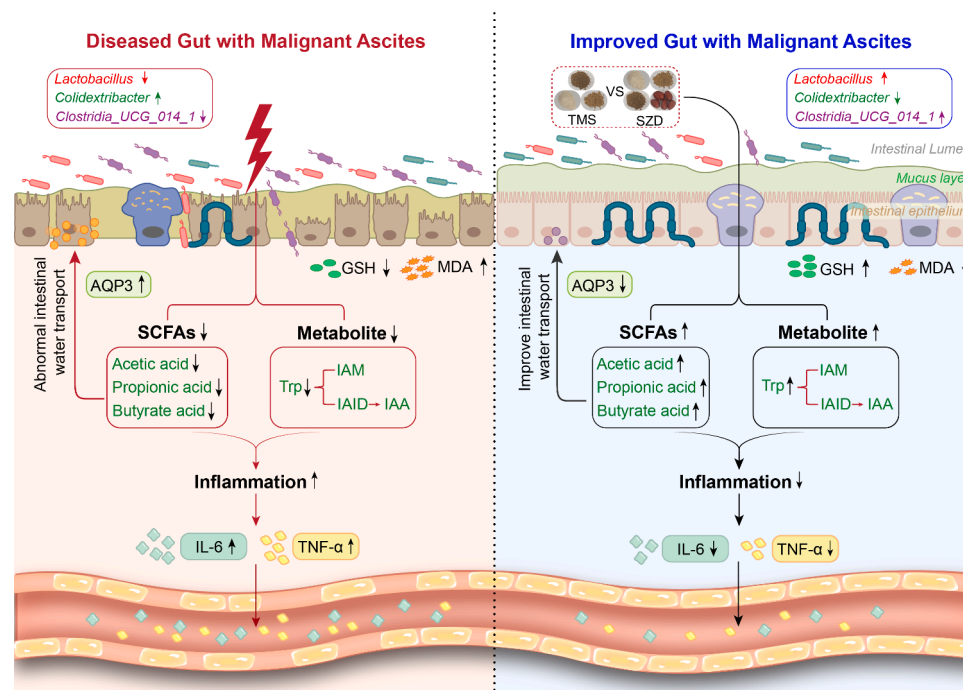


Fig. 12. FJ cooperated with water-expelling members alleviated intestinal injury and malignant ascites through reversing the gut microbial dysbiosis and related metabolic pathways.

valeric acid, propionic acid, l-tryptophan, and l-histidine, but negatively correlated with linoleic acid. In addition, SZD remarkably decreased the relative abundance of *Colidextribacter*, which was negatively correlated with thymidine and positively correlated with 2-hydroxystearic acid, ursodeoxycholic acid, and deoxycholic acid. Tyramine is an important product of tyrosine, which is metabolized by intestinal flora. It could promote the secretion of 5-HT to regulate gastric motility through the enteric nervous system, leading to increasing the fecal water content and accelerating the excretion rate of feces (Dyck et al., 1982; Zhang et al., 2015). Tryptophan was also directly transformed into indole and its derivatives by gut microbiota. These metabolites may combine with aryl hydrocarbon receptors to enhance host immunity and reduce intestinal inflammation (Su et al., 2022; Zhang et al., 2022). *Lactobacillus* could modulate the tryptophan metabolism and block the inhibitory activity of IDO on lymphocytes and natural kill cells, thereby decreasing the invasion of tumor cells (Özçam et al., 2019). Gut microbiota could convert primary bile acids converted into secondary bile acids including deoxycholic acid and ursodeoxycholic acid, which activated protein kinase C/p38 MAPK/NF-κB signaling pathway to inhibit p53 activity. This may lead to an increase in cells with damaged DNA, promoting tumor proliferation (Collins et al., 2023). These clarified that FJ cooperated with water-expelling members alleviated intestinal injury and malignant ascites through reversing the gut microbial dysbiosis and related metabolic pathways (Fig. 12).

Conclusion

In conclusion, FJ cooperated with water-expelling members reduced intestinal injury and alleviated malignant ascites better than water-expelling members alone. The underlying mechanism is related to increasing the relative abundance of *Lactobacillus* and *Clostridia-UCG-014* while decreasing that of *Colidextribacter*. Meanwhile, short-chain fatty acids metabolism and tryptophan metabolism were also modulated. These findings provide a scientific basis for the clinical application of FJ and the safe usage of SZD.

CRediT authorship contribution statement

Shi-Kang Zhou: Investigation, Software, Writing – original draft. **Jin-Di Xu:** Investigation. **Xiao-Qin Gao:** Investigation, Data curation. **Ren-Jie Zhang:** Investigation. **Fang-Fang Cheng:** Formal analysis. **Wei-Feng Yao:** Formal analysis. **Yi Zhang:** Conceptualization, Writing – review & editing. **Ting Geng:** Formal analysis, Writing – review & editing. **Li Zhang:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2024.155895](https://doi.org/10.1016/j.phymed.2024.155895).

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